Parallel lines and transversals

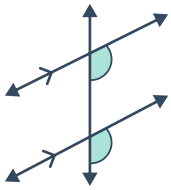
1

a How are parallel lines or segments marked on a diagram?

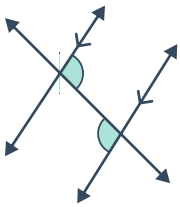
b Draw an example of correctly marked parallel lines.

2 Identify the types of angles formed by the transveral in the following diagrams:

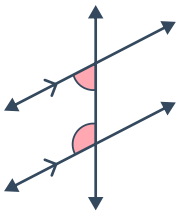
a



b



c



3 State whether the following types of angles are equal, complementary, or supplementary:

a Alternate angles

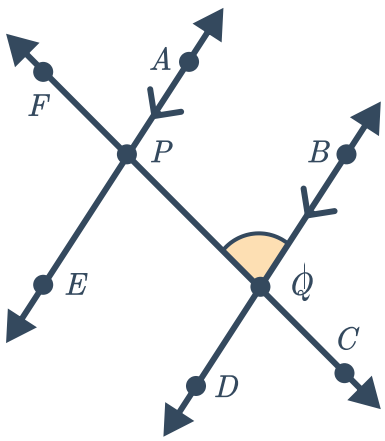
b Cointerior angles

c Corresponding angles

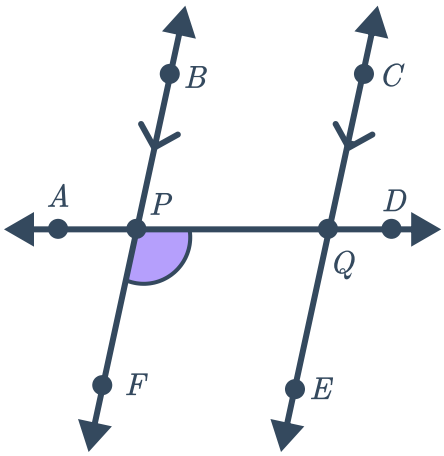
4 For the marked angle in each the following diagrams, name the angle which is:

i Alternate **ii** Cointerior **iii** Corresponding

a



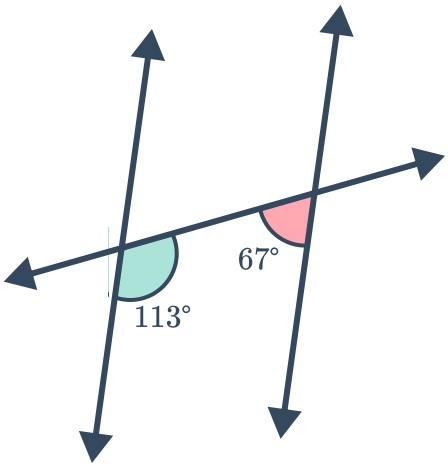
b



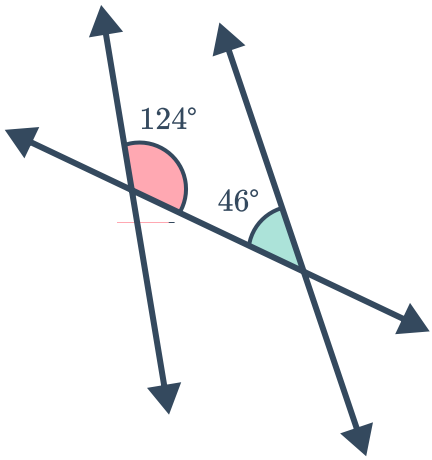
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5 For each of the following diagrams, state whether the diagram contains parallel lines. Explain your answer.

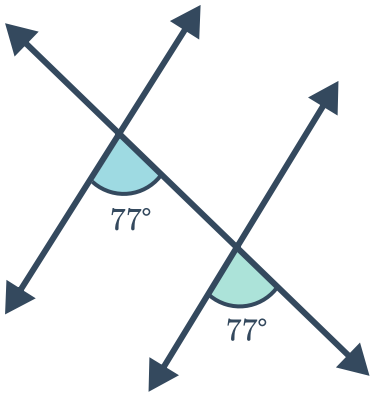
a



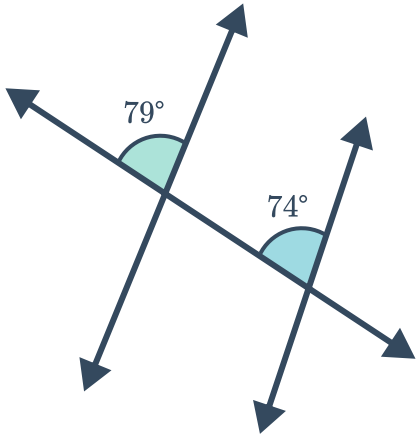
b



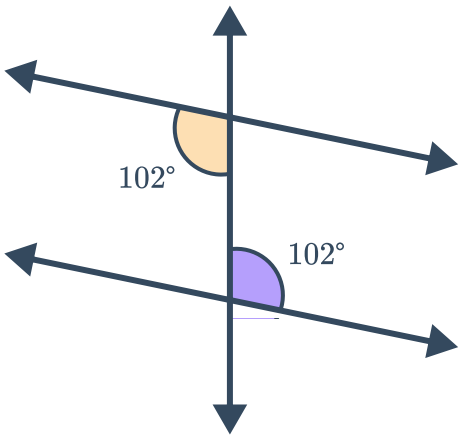
c



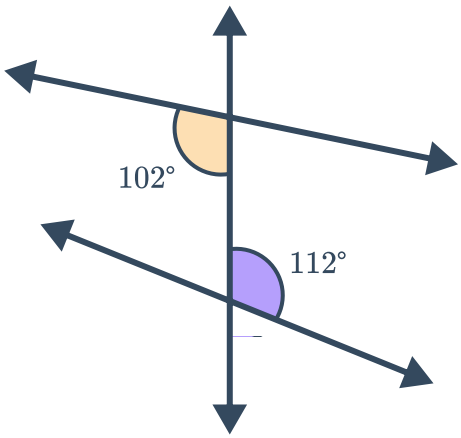
d



e

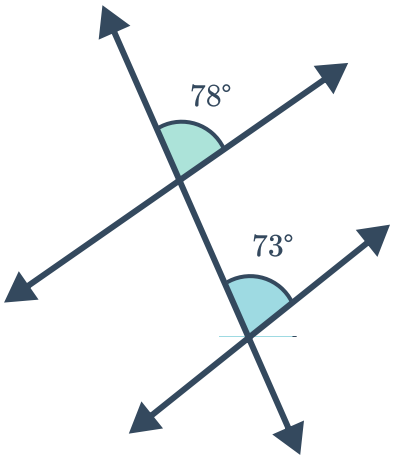


f

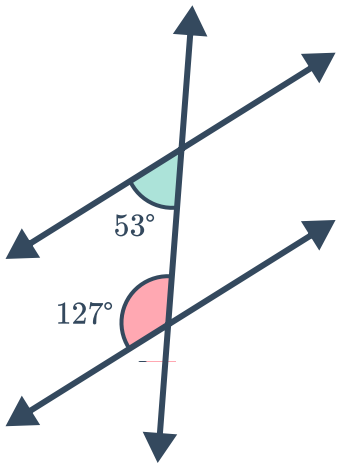


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g

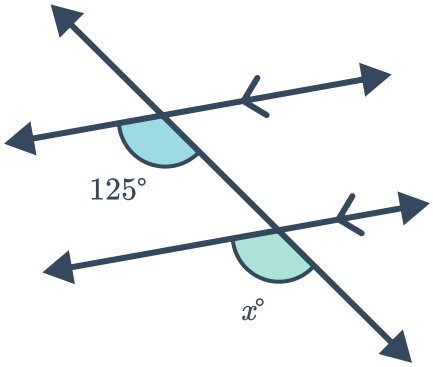


h

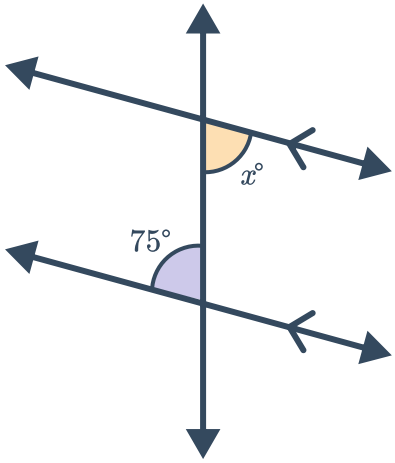


6 For each of the following diagrams, find the value of x:

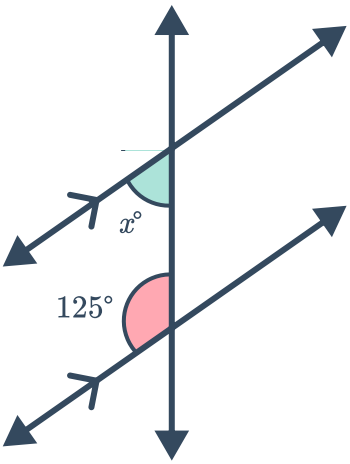
a



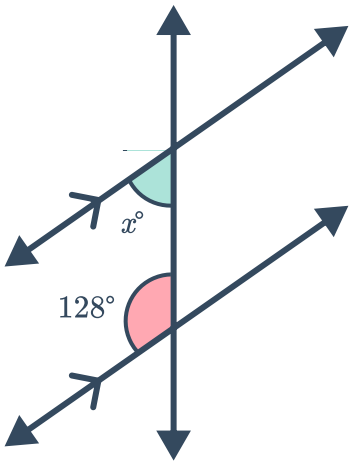
b



c

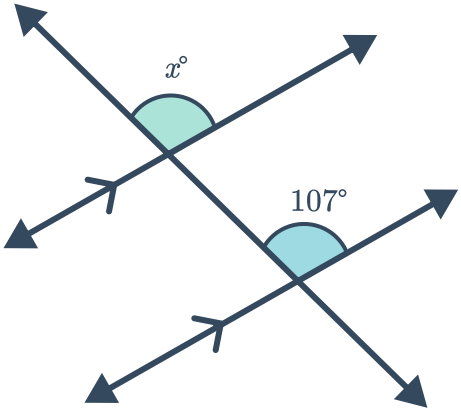


d

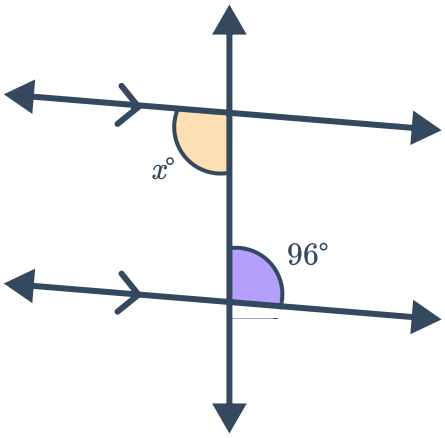


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e



f

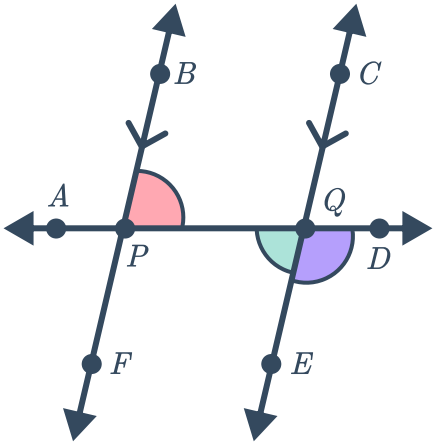


7 Consider the following diagram:

a How are $\angle BPD$ and $\angle AQE$ related?

b How are $\angle AQE$ and $\angle DQE$ related?

c If $\angle BPD = 84^\circ$, find the value of $\angle DQE$.

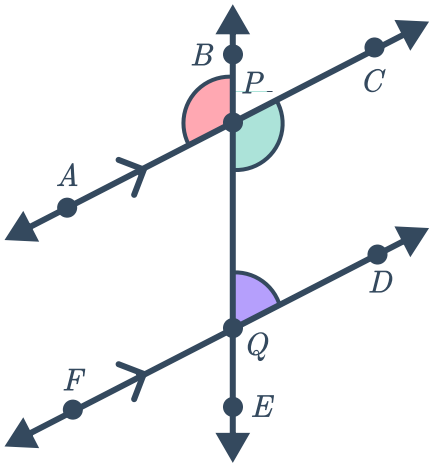


8 Consider the following diagram:

a How are $\angle APB$ and $\angle CPE$ related?

b How are $\angle CPE$ and $\angle BQD$ related?

c If $\angle APB = 107^\circ$, find the value of $\angle BQD$.



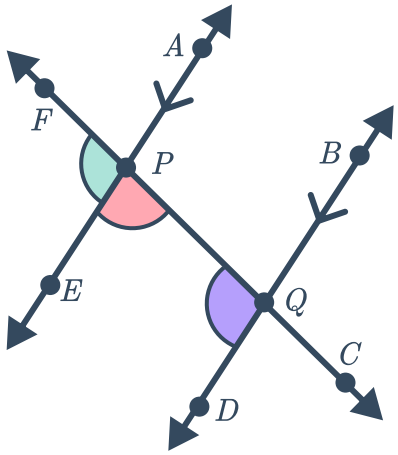
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9 Consider the following diagram:

a How are $\angle DQF$ and $\angle CPE$ related?

b How are $\angle CPE$ and $\angle EPF$ related?

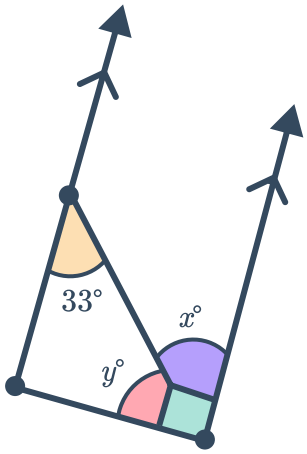
c If $\angle DQF = 97^\circ$, find the value of $\angle EPF$.



10 Consider the following diagram:

a Find the value of x.

b Find the value of y.



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